

## Week 9 Workshop Solutions

Because  $\sigma_t$  is known given  $\mathcal{F}_{t-1}$ ,  $u_t \sim i.i.d.N(0, 1)$  implies that  $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ . Thus we have

$$E(\varepsilon_t|\mathcal{F}_{t-1}) = 0,$$

and

$$V(\varepsilon_t|\mathcal{F}_{t-1}) = \sigma_t^2.$$

1) Apply the Law of Iterated Expectations:

$$E(\varepsilon_t) = E[E(\varepsilon_t|\mathcal{F}_{t-1})].$$

Then, since  $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$ ,

$$E(\varepsilon_t) = E(0) = 0.$$

2) Apply the Law of Total Variance:

$$V(\varepsilon_t) = V[E(\varepsilon_t|\mathcal{F}_{t-1})] + E[V(\varepsilon_t|\mathcal{F}_{t-1})].$$

Since  $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$  and  $V(\varepsilon_t|\mathcal{F}_{t-1}) = \sigma_t^2$ , then

$$V(\varepsilon_t) = V(0) + E(\sigma_t^2).$$

We substitute  $\sigma_t^2$  with  $a_0 + a_1\varepsilon_{t-1}^2$ , so that

$$V(\varepsilon_t) = E(a_0 + a_1\varepsilon_{t-1}^2).$$

Since  $\varepsilon_t$  is stationary, we know that  $E(\varepsilon_{t-1}^2) = E(\varepsilon_t^2)$  and

$$V(\varepsilon_t) = a_0 + a_1E(\varepsilon_t^2).$$

Further, since  $V(\varepsilon_t) = E(\varepsilon_t^2)$ , we have

$$V(\varepsilon_t) = a_0 + a_1V(\varepsilon_t),$$

and solving for  $V(\varepsilon_t)$  gives

$$V(\varepsilon_t) = \frac{a_0}{1 - a_1}.$$

3) From 1) since  $E(\varepsilon_t) = E(\varepsilon_{t-k}) = 0$ , apply the Law of Iterated Expectations

$$\begin{aligned} & E[(\varepsilon_t - E(\varepsilon_t))(\varepsilon_{t-k} - E(\varepsilon_{t-k}))] \\ &= E(\varepsilon_t\varepsilon_{t-k}) \\ &= E[E(\varepsilon_t\varepsilon_{t-k}|\mathcal{F}_{t-1})] \\ &= E[\varepsilon_{t-k}E(\varepsilon_t|\mathcal{F}_{t-1})] \\ &= E[\varepsilon_{t-k} \times 0] \\ &= 0 \end{aligned}$$

for all  $k \geq 1$ . The third equality follows from  $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$ .

These properties imply that

- The unconditional mean is constant for all  $t$ .
- The unconditional variance is constant for all  $t$ .
- $\varepsilon_t$  is uncorrelated over time (no autocorrelation).